wavefunctions do *not* collapse. Wavefunction collapse is an add-on. It was introduced after Schrödinger discovered his equation, in an attempt to account for what experimenters actually see. Whereas a raw, uncollapsed wavefunction embodies the strange idea that a particle is here and there, experimenters never see this. They always find a particle definitely at one location or another; they never see it partially here and partially there; the needle on their measuring devices never hovers in some ghostly mixture of pointing at this value and also at that value.

The same goes, of course, for our own casual observations of the world around us. We never observe a chair to be both here and there; we never observe the moon to be in one part of the night sky as well as another; we never see a cat that is both dead and alive. The notion of wavefunction collapse aligns with our experience by postulating that the act of measurement induces the wavefunction to relinquish quantum limbo and usher one of the many potentialities (particle here, or particle there) into reality.

The Quantum Measurement Puzzle

But how does an experimenter's making a measurement cause a wavefunction to collapse? In fact, does wavefunction collapse really happen, and if it does, what really goes on at the microscopic level? Do any and all measurements cause collapse? When does the collapse happen and how long does it take? Since, according to the Schrödinger equation, wavefunctions do not collapse, what equation takes over in the second stage of quantum evolution, and how does the new equation dethrone Schrödinger's, usurping its usual ironclad power over quantum processes? And, of importance to our current concern with time's arrow, while Schrödinger's equation, the equation that governs the first stage, makes no distinction between forward and backward in time, does the equation for stage two introduce a fundamental asymmetry between time before and time after a measurement is carried out? That is, does quantum mechanics, *including its interface with the world of the everyday via measurements and observations,* introduce an arrow of time into the basic laws of physics? After all, we discussed earlier how the quantum treatment of the past differs from that of classical physics, and by *past* we meant before a particular observation or measurement had